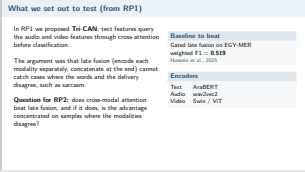
Where the Signal Is

Introduce ourselves. One sentence: in RP1 we proposed a cross-modal attention model for Egyptian Arabic emotion recognition; this is what happened when we built and tested it.

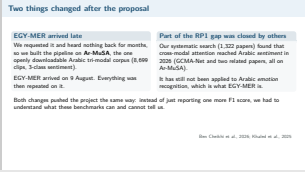
What we set out to test (from RP1)



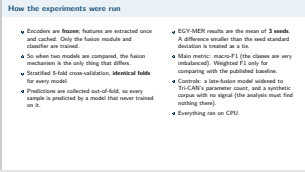
Keep this short, the committee knows RP1. The key sentence is the question at the bottom left: does attention help, and does it help most where the modalities disagree.

Where the Signal Is

Two things changed after the proposal



Say plainly that the RP1 novelty claim was too broad and the updated RP1 report corrects it. Don't be defensive; finding this ourselves is part of doing the review properly.



If asked why frozen: it makes the comparison cleaner, and we did fine-tune the text encoder later to check that the result holds (slide 11).

First result on Ar-MuSA: there was almost nothing to fuse

Ar-MuSA, one seed	macro-F1
Text only	0.546
Audio only	0.418
Visual only	0.433
All three (late fusion)	0.559

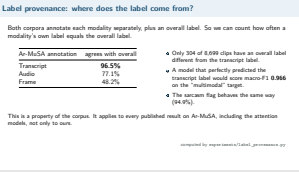
Adding audio and video to text gained only **+0.013**.

Tri-CAN did not beat late fusion either, and neither did two other attention variants.

Before concluding that attention does not work, we checked whether the corpus gives it any room to work in.

This was the point where we could have just written “attention does not help”. Instead we asked why text alone is almost as good as everything together.

Label provenance: where does the label come from?



Label provenance: where does the label come from?

Both corpora annotate each modality separately, plus an overall label. So we can count how often a modality's own label equals the overall label.

Ar-MuSA annotation	agrees with overall
Transcript	**96.5%**
Audio	77.1%
Frame	48.2%

- Only 304 of 8,699 clips have an overall label different from the transcript label.
- A model that perfectly predicted the transcript label would score macro-F1 **0.966** on the "multimodal" target.
- The sarcasm flag behaves the same way (94.9%).

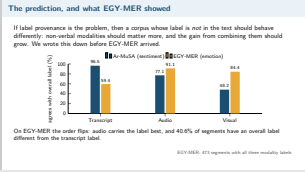
This is a property of the corpus. It applies to every published result on Ar-MuSA, including the attention models, not only to ours.

computed by experiments/label_provenance.py

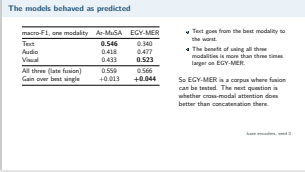
This is the central idea of the project, take your time. The overall label is basically the transcript label. So a text model already has almost all the information and no fusion method has room to show it is better.

Where the Signal Is

└The prediction, and what EGY-MER showed

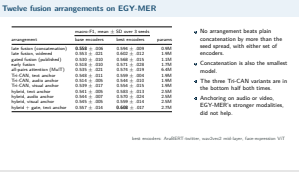


The bars flip. On Ar-MuSA the label lives in the words. On EGY-MER it lives in how things are said and shown.



The prediction from the previous slide holds in the model scores too, not just in the annotations.

Twelve fusion arrangements on EGY-MER



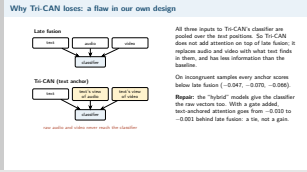
Twelve fusion arrangements on EGY-MER

arrangement	macro-F1, mean ± SD over 3 seeds		
	base encoders	best encoders	params
late fusion (concatenation)	**0.553** ± .026	0.596 ± .009	0.9M
late fusion, validated	0.541 ± .021	0.602 ± .012	1.9M
gated fusion (published)	0.530 ± .010	0.568 ± .015	1.1M
Tri-CAN, full	0.526 ± .021	0.576 ± .018	6.4M
all-pairs attention (MulT)	0.526 ± .021	0.576 ± .018	6.4M
Tri-CAN, text anchor	0.568 ± .011	0.599 ± .004	1.9M
Tri-CAN, audio anchor	0.514 ± .005	0.546 ± .010	1.9M
Tri-CAN, visual anchor	0.539 ± .017	0.556 ± .015	1.9M
hybrid, full	0.541 ± .007	0.570 ± .014	2.5M
hybrid, audio anchor	0.564 ± .007	0.570 ± .014	2.5M
hybrid, visual anchor	0.565 ± .005	0.599 ± .014	2.5M
hybrid + gate, text anchor	0.567 ± .014	**0.608** ± .007	2.5M

- No arrangement beats plain concatenation by more than the seed spread, with either set of encoders.
- Concatenation is also the smallest model.
- The three Tri-CAN variants are in the bottom half both times.
- Anchoring on audio or video, EGY-MER's stronger modalities, did not help.

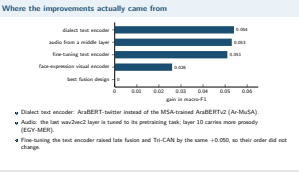
best encoders: AraBERT-twitter; wav2vec2 mid-layer; face-expression ViT

Be careful with wording here. With the best encoders, hybrid+gate is 0.608 against 0.594, but the seed SD is 0.017, so it's a tie, not a win. What we can say: nothing beats concatenation beyond noise, and Tri-CAN itself is never near the top.



We found this ourselves by looking at where Tri-CAN loses. The congruent vs incongruent difference is consistent in direction, but its bootstrap 95% interval is [-0.19, +0.03] and includes zero, so present it as a direction, not a proven effect. The hybrid repair confirms the diagnosis but shows the attended view adds nothing new.

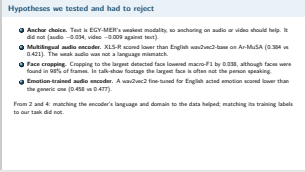
Where the improvements actually came from



Every real gain came from better representations, not from the fusion architecture. The fine-tuning point matters: it shows the fusion result is not just an artefact of frozen encoders.

Where the Signal Is

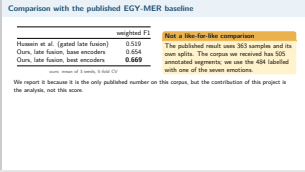
Hypotheses we tested and had to reject



We wrote each of these down before running it. Reporting the ones that failed is part of the result, and it answers "did you try X" questions before they come.

Where the Signal Is

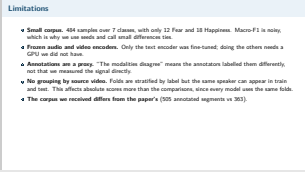
Comparison with the published EGY-MER baseline



If asked: the RP2 report quotes 0.664, which is the single-seed run of the base model. The 3-seed mean is 0.654. Both are well above 0.519, and neither is directly comparable.

Where the Signal Is

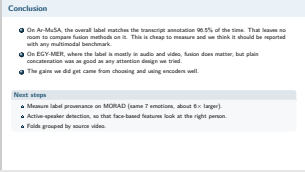
Limitations



Say these before being asked. Each one is also written in the report.

Where the Signal Is

Conclusion



Three points. If time is short, say only the first sentence of each.